

Improvement Roadmap

Our current MVP delivers a solid structural PDF forensics engine that detects metadata anomalies, font inconsistencies, text overlays, fake redactions, and cross-document correlations. Below is a prioritized roadmap of 18 improvements that would elevate Transformatrix from a triage tool into a full enterprise forensics platform.

Improvements	Priority Tiers	Quick Wins
18	3	3

Tier 1 -- High Impact (Accuracy & Coverage Gaps)

#1 P0

OCR + Scanned Document Support

THE GAP:

If a fraudster prints a forged document, scans it back in, and sends a PDF that is just one big image per page, our text/font/overlay detection is completely blind.

THE FIX:

Integrate Tesseract OCR (via pytesseract) to extract text from image-only pages. This unlocks text analysis on scanned documents.
Bonus: Compare OCR-extracted text against any hidden text layer already in the PDF -- mismatches reveal tampering.

#2 P0

Error Level Analysis (ELA) for Images

THE GAP:

We cannot detect if someone Photoshopped a number on a scanned invoice or digitally pasted a signature onto a scanned page.

THE FIX:

Re-compress each page image at a known JPEG quality and compare the error residuals. Edited areas show different compression artifacts, appearing as bright spots on a heatmap.

#3 P1

Font Substitution & Glyph-Level Analysis

THE GAP:

Current font check only looks at font names. A clever fraudster can use Arial on both the original and the forged text -- same name, passes our check.

THE FIX:

Compare actual glyph outlines and font metrics (kerning, ascender/descender ratios) between sections. Two 'Arial' fonts from different sources will have subtly different glyph tables. Uses fonttools (already installed).

#4 P0

Timestamp Forensics (EXIF inside Embedded Images)

THE GAP:

We extract PDF-level metadata but ignore metadata hidden inside embedded JPEG/PNG images (signatures, stamps, logos).

THE FIX:

Use Pillow (already installed) to extract EXIF data from every embedded image. A pasted signature JPEG might still contain 'Software: Adobe Photoshop 2024' or GPS coordinates.

#5 P1

Incremental Update Diffing

THE GAP:

We detect that incremental updates exist but don't show what changed between revisions.

THE FIX:

Use pikepdf to extract each revision separately, then diff the text/images between revisions. Show the user exactly what was changed: 'In revision 2, the date on page 1 was changed from Jan 5 to Jan 15.'

Tier 2 -- Medium Impact (Smarts & Intelligence)

#6 P1

AI/LLM-Powered Contextual Analysis

THE GAP:

The tool flags structural anomalies but cannot understand context. It doesn't know that a company incorporated in Dec 2025 can't have a contract dated Jan 2023.

THE FIX:

Feed the extracted text + findings into a local LLM (or API like Gemini/GPT) to perform semantic cross-referencing. The AI could flag logical impossibilities like date contradictions or name mismatches.

#7 P1

Document Type Classifier

THE GAP:

The tool applies the same heuristics to everything. A marketing brochure generates 86 false positives because it has colorful text.

THE FIX:

Add a lightweight classifier that auto-detects the document type (contract, invoice, bank statement, brochure) and adjusts sensitivity thresholds accordingly.

#8 P2

Copy-Move Forgery Detection (CMFD)

THE GAP:

We detect reused images across files but not within a single page (e.g. someone copying a cell in a table and pasting it over another cell).

THE FIX:

Implement block-matching algorithms (SIFT/SURF keypoint matching) on each page to find duplicated regions within the same image. Requires opencv-python.

#9 P2

Digital Signature Validation

THE GAP:

We detect that a /Sig object exists but can't verify if the cryptographic signature is valid, revoked, or covers the whole file.

THE FIX:

Integrate pyhanko (Python PDF signing/verification library) to verify the certificate chain, check revocation status (CRL/OCSP), and confirm the signature covers all bytes.

#10 P2

Noise & Compression Artifact Analysis

THE GAP:

Beyond ELA, there are other statistical tests for image tampering we are not running.

THE FIX:

Implement noise-level estimation per image block. Forged regions will have a different noise profile than the rest of the page because they were processed separately.

Tier 3 -- Polish & Enterprise Features

#11 P2

Historical Signature/Stamp Database

THE GAP:

Cross-document matching only works within the current upload session.

THE FIX:

Build a persistent local database (SQLite) that stores perceptual hashes of every signature/stamp ever analyzed. Alert: 'This exact signature was previously seen in case #XYZ from 2023.'

#12 P2

PDF Report Enhancements

THE GAP:

The current PDF report is text-only with no visual evidence.

THE FIX:

Embed actual screenshot crops of flagged regions into the PDF report. Add a risk score gauge, timeline visualization of metadata dates vs. claimed dates, and side-by-side signature comparison views.

#13 P2

Multi-Format Support (.docx, .xlsx, .pptx)

THE GAP:

Only PDFs are supported. Word and Excel files have even richer forensic metadata.

THE FIX:

Extend to analyze .docx, .xlsx, and .pptx files directly using python-docx and openpyxl. These formats have revision history, tracked changes, and embedded OLE objects.

#14 P3

Batch Processing & CLI Mode

THE GAP:

You can only use it through the Streamlit web UI.

THE FIX:

Add a CLI entry point (`python forensics_engine.py --input ./docs/ --output report.pdf`) for batch processing hundreds of files in a pipeline without opening a browser.

#15 P3

Redaction Verification

THE GAP:

We detect fake redactions but don't verify if real redactions were done properly.

THE FIX:

For properly redacted documents, verify that the underlying text layer has actually been removed (not just visually hidden). Many 'professional' redaction tools only draw a black rectangle.

#16 P3

Watermark & Background Analysis

THE GAP:

We don't check for tampered or removed watermarks.

THE FIX:

Detect and extract watermark layers. Flag if a watermark appears inconsistently (present on some pages, missing on others) or if the watermark text doesn't match the document's claimed source.

#17 P3

Audit Trail & Chain of Custody

THE GAP:

No record of who uploaded what and when.

THE FIX:

Add user authentication, log every upload with timestamps and file hashes, and include this audit trail in the exported report. Critical for legal admissibility.

#18 P3

API / Webhook Integration

THE GAP:

No way to integrate with existing compliance or case management systems.

THE FIX:

Wrap the forensics engine in a FastAPI service with webhook callbacks. Compliance systems can POST a PDF and receive a JSON findings report back programmatically.

Priority Matrix

Improvement	Accuracy	Impact	Effort	Priority	Status
OCR + Scanned Docs	Critical	Medium	P0	Planned	
Error Level Analysis (ELA)	High	Medium	P0	Planned	
EXIF from Embedded Images	High	Low	P0	Planned	
Font Glyph Analysis	Medium	High	P1	Planned	
Incremental Update Diffing	High	Medium	P1	Planned	
AI/LLM Context Analysis	High	Medium	P1	Planned	
Document Type Classifier	Medium	Medium	P1	Planned	
Copy-Move Forgery (CMFD)	Medium	High	P2	Backlog	
Digital Signature Validation	Medium	Medium	P2	Backlog	
Noise Artifact Analysis	Medium	High	P2	Backlog	
Historical Signature DB	Medium	Low	P2	Backlog	
PDF Report Enhancements	Low	Medium	P2	Backlog	
Multi-Format (.docx/.xlsx)	High	High	P2	Backlog	
Batch CLI Mode	Low	Low	P3	Backlog	
Redaction Verification	Medium	Medium	P3	Backlog	
Watermark Analysis	Low	Medium	P3	Backlog	
Audit Trail	Low	Medium	P3	Backlog	
API / Webhook	Low	Medium	P3	Backlog	

Recommended Next Steps

Start with EXIF extraction (lowest effort, high payoff), then OCR for scanned documents, then ELA image forensics. These three alone would cover the biggest blind spots in the current MVP and deliver immediate value to clients.